DataCite Corpus - Data Exploration & Descriptive Statistics

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DataCite Corpus - Exploratory Analysis

```
## FUNCTION: exp_summary - argument dat (dataframe)
## Argument dat takes a dataframe. Column names are assumed to
## match the columns provided in the DataCite Corpus as of
## 11 October 2026. Documentation and file downloads available
## at https://zenodo.org/records/16901115

get_dc_summary <- function(filename) {
  ## Returns descriptive statistics of the data file - number
  ## of rows, number & proportion of rows with publication DOIs,
  ## number & proportion of rows with a 'created' date,
  ## number & proportion of rows with publication dates,
  ## number & proportion of rows with a funder listed.

  dat <- read_csv(file = filename, na = c("", "NA", "none"))

  summarylist <- dat |> summarise(
    n = nrow(dat),
    n_doi = str_detect(dat$publication, pattern = "doi") |> sum(na.rm = TRUE),
    n_created_date = sum(1 - is.na(dat$created), na.rm = TRUE),
    n_pub_date = sum(1 - is.na(dat$publishedDate), na.rm = TRUE),
    n_funders = sum(1 - is.na(dat$funders), na.rm = TRUE),
    n_nih = str_detect(dat$funders, pattern = "(NIH)|(National Institutes of Health)") |>
      sum(na.rm = TRUE)
  ) |> unlist()

  return(summarylist)
}


# Function computing descriptive stats

summary_datacite <- function(filenames) {
  cl <- makeCluster(8)
  registerDoParallel(cl)

  summ_dc <- foreach(
    fn = filenames,
    .combine = "rbind",
    .export = "get_dc_summary",
    .packages = "tidyverse"
```

```
  ) %dopar% {
    get_dc_summary(fn)
  }

  stopCluster(cl)

  return(summ_dc)
}
```

```
## TODO: Modify dirpath to current directory
dirpath = "/Users/ashvin/Desktop/biomed-retractions/datacite"

# Gets the names of files to be read int, then calls rw_merge.
files <- list.files(path = paste0(dirpath, "/data"),
                    full.names = TRUE, recursive = TRUE)



summary_dat <- summary_datacite(files) |> as.data.frame()
row.names(summary_dat) <- c(1:nrow(summary_dat))


summary_dat |> summarise(
  N.obs = sum(n),
  N.doi = sum(n_doi),
  Prop.doi = sum(n_doi) / sum(n),
  N.cr.date = sum(n_created_date),
  Prop.cr.date = sum(n_created_date) / sum(n),
  N.funders = sum(n_funders),
  Prop.funders = sum(n_funders) / sum(n),
  N.NIH = sum(n_nih),
  Prop.NIH = sum(n_nih) / sum(n)
) |>
  t() |>
  kable()
```

N.obs	1.069774e+07
N.doi	1.069629e+07
Prop.doi	9.998643e-01
N.cr.date	1.069774e+07
Prop.cr.date	1.000000e+00
N.funders	9.720000e+04
Prop.funders	9.086000e-03
N.NIH	5.202000e+03
Prop.NIH	4.863000e-04

```
df <- read_csv("merged.csv")
```

```
## Rows: 14970 Columns: 21
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr  (17): data.record.id, data.repository, data.title, data.id, data.citati...
## dbl   (3): pub.record.id, pub.year, pub.retraction.year
```

```
## dttm  (1): data.created.date
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
df |> summarise(
  N.obs = nrow(df),
  N.doi = sum(!is.na(data.citation.id)),
  Prop.doi = sum(!is.na(data.citation.id)) / nrow(df),
  N.cr.date = sum(!is.na(data.created.date)),
  Prop.cr.date = sum(!is.na(data.created.date)) / nrow(df),
  N.pub.date = sum(!is.na(pub.date)),
  N.funders = sum(!is.na(pub.date)) / nrow(df),
  Prop.funders = sum(!is.na(data.funders)) / nrow(df),
  N.NIH = sum(str_detect(data.funders, pattern = "(NIH)|(NATIONAL INSTITUTES OF HEALTH)") |
                str_detect(data.funders.ror, pattern = "(NIH)|(NATIONAL INSTITUTES OF HEALTH)"),
              na.rm = TRUE),
  Prop.NIH = sum(str_detect(data.funders, pattern = "(NIH)|(NATIONAL INSTITUTES OF HEALTH)") |
                str_detect(data.funders.ror, pattern = "(NIH)|(NATIONAL INSTITUTES OF HEALTH)"),
                na.rm = TRUE) /
    nrow(df)
)  |>
  t() |>
  kable()
```

N.obs	1.497e+04
N.doi	1.497e+04
Prop.doi	1.000e+00
N.cr.date	1.497e+04
Prop.cr.date	1.000e+00
N.pub.date	1.497e+04
N.funders	1.000e+00
Prop.funders	1.002e-03
N.NIH	4.000e+00
Prop.NIH	2.672e-04

Only 1 NIH-funded publication that also appears in RW Database

```
df |> filter(!is.na(data.funders) | !is.na(data.funders.ror)) |>
  select(data.title, pub.title, pub.journal, data.funders, data.funders.ror)
```

```
## # A tibble: 15 x 5
##    data.title                pub.title pub.journal data.funders data.funders.ror
##    <chr>                     <chr>     <chr>       <chr>        <chr>
##  1 "PITTSBURGH YOUTH STUDY ~ Chronic ~ Psychology~ PEW CHARITA~ PEW CHARITABLE ~
##  2 "<B>RETRACTION:</B>SUPPO~ Dissecti~ Gigascience NATIONAL RE~ NONE NONE; NONE~
##  3 "<B>RETRACTION:</B>SUPPO~ Dissecti~ Gigascience NATIONAL RE~ NONE NONE; NONE~
##  4 "SECOND-GENERATION CITAT~ Effects ~ Chest       ALFRED P. S~ ALFRED P. SLOAN~
##  5 "PITTSBURGH YOUTH STUDY ~ Chronic ~ Psychology~ UNITED STAT~ NONE NONE; PEW ~
##  6 "HYPOXIA PRECONDITIONED ~ Hypoxia ~ PLoS One    NATIONAL NA~ NATIONAL NATURA~
##  7 "HINC DATASET FROM: HOME~ Homeopat~ The Oncolo~ INSTITUTE F~ NONE NONE; NONE~
##  8 "HINC DATASET FROM: HOME~ Homeopat~ The Oncolo~ INSTITUTE F~ NONE NONE; NONE~
##  9 "DATA REPOSITORY FOR AUT~ Precise ~ Nature Com~ EU FUNDING ~ NONE NONE; NONE~
## 10 "PITTSBURGH YOUTH STUDY ~ Chronic ~ Psychology~ UNITED STAT~ NATIONAL INSTIT~
```

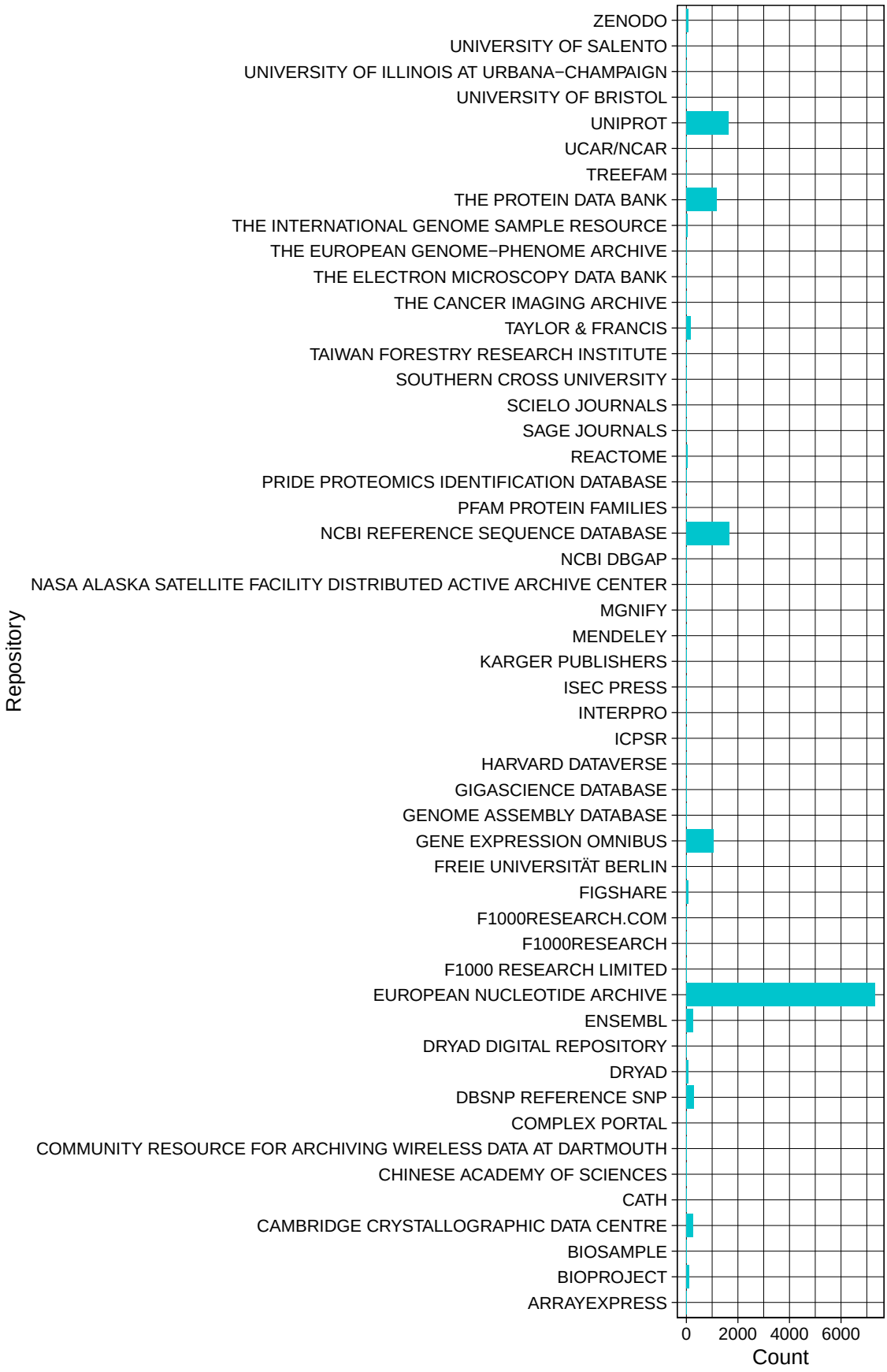
```
## 11 "EUSPEECH: A NEW DATASET~ Liberals~ PLoS One    HORIZON 2020 NONE NONE
## 12 "LATINO NATIONAL SURVEY ~ Tuning I~ American J~ IRVINE FOUN~ NONE NONE; NONE~
## 13 "PITTSBURGH YOUTH STUDY ~ Chronic ~ Psychology~ UNITED STAT~ NONE NONE; NONE~
## 14 "MONOCENTRIC RETROSPECTI~ Outcomes~ New Microb~ AGENCE NATI~ AGENCE NATIONAL~
## 15 "INVESTIGATION RESULTS O~ Niches o~ Ecology an~ NATIONAL NA~ NATIONAL NATURA~
```

Number of repositories

```
df |> count(data.repository) |> arrange(desc(n))
```

```
## # A tibble: 53 x 2
##    data.repository                              n
##    <chr>                                    <int>
##  1 EUROPEAN NUCLEOTIDE ARCHIVE               7316
##  2 NCBI REFERENCE SEQUENCE DATABASE          1655
##  3 UNIPROT                                   1641
##  4 THE PROTEIN DATA BANK                     1178
##  5 GENE EXPRESSION OMNIBUS (GEO)             1052
##  6 <NA>                                       519
##  7 DBSNP REFERENCE SNP                        294
##  8 ENSEMBL                                    273
##  9 CAMBRIDGE CRYSTALLOGRAPHIC DATA CENTRE     265
## 10 TAYLOR & FRANCIS                           177
## # i 43 more rows
```

```
df |>
  filter(!is.na(data.repository) & data.repository != "UNPUBLISHED") |>
  mutate(
      data.repository.mod = str_remove(data.repository, pattern = "\\s\\(.*\\)")
    ) |>
  mutate(
      data.repository.mod = str_remove(data.repository.mod, pattern = "\\s-\\s.*")
    ) |>
  ggplot(aes(data.repository.mod)) +
    geom_bar(fill = "turquoise3") +
    labs(x = "Repository", y = "Count") +
    coord_flip() +
    theme_linedraw()
```


Repository
ZENODO
UNIVERSITY OF SALENTO
UNIVERSITY OF ILLINOIS AT URBANA−CHAMPAIGN
UNIVERSITY OF BRISTOL
UNIPROT
UCAR/NCAR
TREEFAM
THE PROTEIN DATA BANK
THE INTERNATIONAL GENOME SAMPLE RESOURCE
THE EUROPEAN GENOME−PHENOME ARCHIVE
THE ELECTRON MICROSCOPY DATA BANK
THE CANCER IMAGING ARCHIVE
TAYLOR & FRANCIS
TAIWAN FORESTRY RESEARCH INSTITUTE
SOUTHERN CROSS UNIVERSITY
SCIELO JOURNALS
SAGE JOURNALS
REACTOME
PRIDE PROTEOMICS IDENTIFICATION DATABASE
PFAM PROTEIN FAMILIES
NCBI REFERENCE SEQUENCE DATABASE
NCBI DBGAP
NASA ALASKA SATELLITE FACILITY DISTRIBUTED ACTIVE ARCHIVE CENTER
MGNIFY
MENDELEY
KARGER PUBLISHERS
ISEC PRESS
INTERPRO
ICPSR
HARVARD DATAVERSE
GIGASCIENCE DATABASE
GENOME ASSEMBLY DATABASE
GENE EXPRESSION OMNIBUS
FREIE UNIVERSITÄT BERLIN
FIGSHARE
F1000RESEARCH.COM
F1000RESEARCH
F1000 RESEARCH LIMITED
EUROPEAN NUCLEOTIDE ARCHIVE
ENSEMBL
DRYAD DIGITAL REPOSITORY
DRYAD
DBSNP REFERENCE SNP
COMPLEX PORTAL
COMMUNITY RESOURCE FOR ARCHIVING WIRELESS DATA AT DARTMOUTH
CHINESE ACADEMY OF SCIENCES
CATH
CAMBRIDGE CRYSTALLOGRAPHIC DATA CENTRE
BIOSAMPLE
BIOPROJECT
ARRAYEXPRESS
0
2000
4000
6000
Count